

FINITE INFORMATION AND INFINITE CONTINUATION

What established mathematics already solves, what it does not,
and a theorem-first research program beyond continued fractions

Principal verdict

The qualitative thesis is already a standard consequence of continuous symbolic coding: finite prefixes define nested continuation sets; continuity localizes those sets; the admissible language fixes the asymptotic reachable set. Continued-fraction Experiments 1 and 2 instantiate this established structure. The remaining research opportunity is quantitative and conditional: identify coding-invariant resource laws, or prove precisely how target geometry controls minimum containment cost.

Scientific synthesis and novelty audit

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Abstract

This paper reconstructs the research question behind three continued-fraction experiments: whether finite organization governs, restricts, and localizes infinite continuation. It then asks what established mathematics already proves across topology, symbolic dynamics, automata theory, iterated function systems, information theory, ergodic theory, domain theory, predictive-state modeling, number systems, and symbolic control. The central result is a synthesis theorem, not a new theorem: if an infinite object is represented by a compact symbolic space and a continuous coding map, every finite prefix defines a nested continuation set, and the diameters of those sets shrink uniformly. An open target contains a complete continuation set exactly when it intersects the coded reachable set. Restricting the alphabet or admissible language changes the reachable set; increasing depth only refines localization within it. This theorem subsumes the qualitative conclusions of Continued-Fraction Experiments 1 and 2. Their exact finite tables remain valid computational certificates, but they do not establish new foundational mathematics. The proposed Experiment 3 quantities are then audited. Minimum raw depth and alphabet size are not intrinsic: block recoding changes them without changing the represented object. Geometric overshoot is more defensible, but remains coordinate- and target-dependent. The paper therefore terminates the broad experimental program and replaces it with a theorem-first program. The strongest immediate candidate is a target-clearance law relating minimum geometric overshoot to the largest attractor-centered ball contained in a target. A small certified experiment is specified only as a conjecture discriminator and proof-development instrument.

Scope and evidentiary standard

The review distinguishes four evidentiary levels: peer-reviewed theorem, established textbook synthesis, project computation, and candidate open problem. A quantity is not treated as novel merely because its notation or control-theoretic interpretation is new. A problem is called an open candidate only when the reviewed literature does not provide an equivalent theorem; that label is not a substitute for a specialist MathSciNet or Zentralblatt novelty search.

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1. Executive findings

Bottom line

The broad question is solved under explicit assumptions. The correct scientific statement is conditional, not metaphysical: in a compact symbolic coding with a continuous decoder, finite prefixes localize the set of compatible infinite continuations. The admissible language determines the asymptotic possibility set. Entropy, contraction, Lyapunov exponents, and dimension quantify the rate and residual freedom.

1. Experiment 1 reproduced continued-fraction cylinder and continuant theory. Finite prefixes define exact cylinders; convergent denominators determine width; digit order affects continuants and therefore width.
2. Experiment 2 reproduced restricted-digit continued-fraction Cantor-set theory. A bounded alphabet fixes a compact fractal reachable set with permanent complementary gaps. Greater depth resolves the set but does not enlarge it.
3. The project computations are still useful. They verify the implementation, give exact finite-resolution counts, expose greedy failures, and provide reproducible examples. Their scientific role is validation and illustration, not foundational discovery.
4. The cross-disciplinary principle is older and broader than continued fractions. It appears as cylinder sets in symbolic dynamics, right-congruence classes in automata, cells in iterated function systems, finite approximants in domain theory, causal states in computational mechanics, and finite abstractions in control.
5. The principle is not universally valid without assumptions. Discontinuous codes, noncompact spaces, expanding forward dynamics, and Turing-complete systems defeat uniform localization, computability, or decidability.
6. Experiment 3 should not be run in its present large-grid form. Raw depth and alphabet ceiling are representation-dependent. A block code can divide depth by two and square the alphabet without changing the represented set or outcomes.
7. The defensible research opportunity is to identify normalized or geometric costs that are stable under a declared class of recodings, then prove target-conditioned scaling laws. Computation should test a precise conjecture, not discover a vocabulary.

1.1 Status matrix

Question	Status	Established framework	Action
Finite prefix defines compatible continuation set	Solved	Cylinder sets / prefix topology	No experiment
Longer prefix localizes under continuous compact coding	Solved	Uniform continuity on compact symbolic space	No experiment
Restricted alphabet/language fixes reachable set	Solved	Subshifts, IFS attractors, automata languages	No experiment
Permanent gaps for bounded continued-fraction digits	Solved	Continued-fraction Cantor sets	No experiment
Order changes continued-fraction focusing	Solved as phenomenon	Continuants and rearrangement theory	No broad experiment
Raw minimum depth as intrinsic cost	False	Destroyed by block recoding	Do not use unnormalized
Alphabet ceiling as intrinsic operator strength	False in general	Alphabet is a representation choice	Do not generalize
Geometric overshoot versus target clearance	Not clearly settled in exact target-conditioned form	Moran covers, bounded distortion, stopping sets	Focused theorem audit + pilot
Complexity of exact target containment for a fixed Möbius IFS	Candidate	Automata/IFS decision and complexity theory	Only after formal problem statement
Universal algorithm across recursive systems	Impossible	Undecidability in universal computation	Terminate claim

1.2 Recommended disposition of current work

- Retain Experiments 1 and 2 as a verified computational appendix and software-validation suite.
- Rewrite the central paper as a synthesis of established mathematics, not as a discovery paper about continued fractions.
- Cancel the full 130,798-cell Experiment 3 run until a recoding-invariant quantity and theorem-scale conjecture are fixed.
- Run one small certified pilot only after the target-clearance conjecture survives a specialist literature audit.

2. The actual research question

The project record states the intended destination clearly: continued fractions are a laboratory, not the final subject. The desired theory concerns reachability, control cost, localization, and universality across recursive systems. In the project's own formulation, the central hypothesis is:

finite organization governs, restricts, and localizes infinite continuation.

To evaluate this scientifically, each term must be replaced by a mathematical object. Otherwise the statement is too elastic to falsify.

2.1 Operational translation

Project term	Mathematical replacement
Finite organization	A finite word, finite state, finite automaton state, finite partition cell, or finite approximation.
Infinite continuation	An infinite tail, path, orbit, stream, function, or directed limit compatible with the finite object.
Governs	Changes the admissible continuation set through a specified transition or coding law.
Restricts	Replaces the full space of futures by a subset determined by the prefix, state, or language.
Localizes	Reduces metric diameter, uncertainty, measure, entropy, or another declared size of the continuation set.
Control cost	The minimum declared resource needed to force a target: symbols, information, geometric contraction, computation, energy, or time.
Universality	The same theorem, with the same assumptions and invariant quantities, applies across non-isomorphic model classes.

2.2 The question separates into three levels

1. Qualitative topology: Do finite prefixes define nested neighborhoods or continuation sets? This is solved.
2. Quantitative geometry and information: How fast do those sets shrink, and how much residual freedom remains? This is solved in many model classes through contraction, entropy, pressure, dimension, and Lyapunov theory.
3. Algorithmic control: What is the cheapest certificate for a prescribed target, and can it be computed? This is model-dependent. Some cases are decidable and efficiently approximable; universal cases are undecidable.

2.3 Why the distinction matters

A broad qualitative statement can be true for a trivial representational reason. The Hausdorff-Alexandroff theorem says that every nonempty compact metric space is a continuous image of Cantor space. Therefore almost any compact object can be encoded by an infinite binary sequence whose prefixes progressively localize its image. This establishes breadth, but it also creates a danger: if the coding map is arbitrary, the result may describe the representation rather than the object. A genuine theory must identify which quantities survive permissible changes of coding.

3. General mathematical model

3.1 Symbolic source and coding map

Let Σ be a finite alphabet. Let $\Sigma^{\mathbb{N}}$ be the space of one-sided infinite sequences equipped with the prefix metric. Let Y be a closed set of admissible sequences, such as a full shift, subshift, automaton language, or constrained path space. Let X be a compact metric state space, and let $\pi: Y \rightarrow X$ be a continuous coding map.

$A = \pi(Y)$ is the asymptotic reachable set.

For a finite word w of length n , define the prefix cylinder and its coded continuation set:

$$[w]_Y = \{s \in Y : s_1 \dots s_n = w\}, \quad C_w = \pi([w]_Y).$$

The object C_w is the exact mathematical version of “all infinite futures still compatible with the finite organization w .”

3.2 Core quantities

Quantity	Definition and warning
Reachability	An open target U is robustly reachable if some nonempty C_w is contained in U .
Localization	$\text{diam}(C_w)$, or a measure/information analogue, quantifies remaining uncertainty.
Minimum depth	$N^*(U) = \min\{ w : C_w \subseteq U\}$; meaningful only after the coding is fixed.
Geometric overshoot	$O^*(U) = \inf \log(\text{diam}(U)/\text{diam}(C_w))$ over nonempty $C_w \subseteq U$.
Residual freedom	Cardinality, topological entropy, measure, conditional entropy, or dimension of $[w]_Y$ or C_w .
Language/alphabet resource	A nested family $Y_K \subseteq Y_{K+1}$; its image $A_K = \pi(Y_K)$ fixes possibilities at resource K .

3.3 Three different processes that must not be conflated

- Generative refinement: append symbols to choose a more specific object. Prefixes usually localize by construction.
- Forward prediction: evolve a state through time. In chaotic or unstable dynamics, finite uncertainty can expand rather than shrink.
- Control synthesis: choose inputs to force a target despite dynamics and disturbances. Reachability depends on controllability and stability, not only on coding.

The continued-fraction experiments study generative refinement. Their conclusions cannot be transferred directly to forward physical causation or general control systems without additional assumptions.

4. The standard coding theorem

Synthesis theorem - established consequence, not a new theorem

Compact symbolic source + continuous coding is sufficient for finite-prefix localization and exact open-target reachability. This is the mathematical core the project was experimentally rediscovering.

4.1 Theorem

Let Σ be finite, let $Y \subseteq \Sigma^{\mathbb{N}}$ be compact, let X be a metric space, and let $\pi: Y \rightarrow X$ be continuous. For each admissible word w , let $C_w = \pi([w]_Y)$, and let $A = \pi(Y)$. Then:

1. Nestedness: if wa is an admissible extension of w , then $C_{wa} \subseteq C_w$.
2. Uniform localization: the maximum diameter of a nonempty length- n continuation set tends to zero as n tends to infinity.
3. Robust reachability: for every open $U \subseteq X$, there exists a nonempty prefix w with $C_w \subseteq U$ if and only if $U \cap A$ is nonempty.

4. Finite target-mask saturation: for any finite collection of open targets, every reachable target has a certificate by some common finite depth N .
5. Resource monotonicity: if $Y_K \subseteq Y_{K+1}$, then $A_K \subseteq A_{K+1}$; increasing depth refines continuation sets within A_K , while changing K changes the asymptotic reachable set.

4.2 Proof

Nestedness follows because $[wa]_Y \subseteq [w]_Y$. For localization, a length- n prefix cylinder has diameter at most 2^{-n} in the prefix metric. Continuity on compact Y implies uniform continuity, so images of sufficiently small prefix cylinders have uniformly small diameter. For reachability, if $C_w \subseteq U$ then U intersects A . Conversely, choose $x \in U \cap A$ and a sequence $s \in Y$ with $\pi(s) = x$. Continuity gives a prefix neighborhood of s whose image lies inside U . For a finite target collection, take the maximum of the finitely many first-success depths. Resource monotonicity follows directly from set inclusion.

4.3 Quantitative rate under Hölder or contracting coding

If the coding satisfies a Hölder bound whenever two sequences agree for n symbols,

$$d_X(\pi(s), \pi(t)) \leq L \cdot 2^{-(\alpha n)},$$

then every length- n continuation set satisfies $\text{diam}(C_w) \leq L \cdot 2^{-(\alpha n)}$. If $x = \pi(s)$ lies at distance δ from the complement of an open target U , then any prefix length satisfying

$$n \geq \log_2(L/\delta) / \alpha$$

is sufficient for containment. In self-similar and conformal iterated function systems, products of contraction ratios or derivatives replace the uniform $2^{-\alpha n}$ rate. In regular continued fractions, convergent denominators give the exact nonuniform rate.

4.4 What the theorem does and does not say

Claim boundary	Meaning
It does say	Finite prefixes localize under the declared compact continuous coding.
It does say	The admissible language fixes the exact asymptotic possibility set.
It does say	Every open target intersecting that set contains some full continuation cylinder.
It does not say	The coding is unique, natural, physical, or causally fundamental.
It does not say	Raw prefix length is an intrinsic amount of organization.
It does not say	Finite observations predict unstable forward dynamics.
It does not say	Reachability or minimum cost is algorithmically decidable in every recursive system.

5. What each established field already contributes

5.1 Symbolic dynamics

Morse and Hedlund introduced symbolic dynamics as the study of shift spaces and symbolic trajectories. Modern symbolic dynamics formalizes finite words as cylinder sets, admissibility through subshifts, and long-run complexity through topological entropy. The project's "finite prefix plus infinite continuation" is standard symbolic coding. A subshift's language determines which finite words and infinite tails are possible; depth selects a cylinder but does not alter the shift space itself [1,2].

5.2 Automata and formal languages

The Myhill-Nerode framework identifies two finite histories when they admit exactly the same future continuations. A language is regular precisely when there are finitely many such continuation-equivalence classes. Thus the project's finite state as compressed history, constraint, and transition operator is classical automata theory. Büchi automata extend the idea to

infinite words and ω -languages. The relevant established lesson is stronger than the project's qualitative claim: finite continuation structure is exactly what finite automata recognize [3-5].

5.3 Iterated function systems and graph-directed constructions

Hutchinson proved the existence and uniqueness of compact attractors for finite contracting systems. A finite operator word composes contractions and maps the attractor into a nested cell. Mauldin and Williams generalized the construction to graph-directed systems, where an admissible path controls both continuation and geometry. Pressure, transfer operators, and bounded-distortion theory quantify dimension and cylinder scales [6-8]. Continued-fraction restricted-digit sets are conformal IFS limit sets.

5.4 Information and ergodic theory

Shannon entropy measures the average information carried by symbols and codewords. The Shannon-McMillan-Breiman theorem states that for a stationary ergodic source, typical length- n cylinders have probability approximately $\exp(-nh)$, where h is the entropy rate. This supplies a principled replacement for raw depth: information content $-\log \mu([w])$ rather than the number of symbols. In smooth dynamical systems, Lyapunov exponents quantify exponential expansion or contraction; in many conformal settings, entropy divided by a Lyapunov exponent determines dimension or typical geometric scaling [9-13].

5.5 Domain theory and computable analysis

Scott's domain theory models infinite computational objects as limits of directed systems of finite approximations. Compact or finite elements represent finite information; the infinite object is the supremum of increasingly informative approximants. Computable analysis similarly represents real numbers and functions through converging names. This is an exact established mathematics of finite information and infinite completion, but it concerns representation and computation, not a universal physical ontology [14,15].

5.6 Predictive-state and computational-mechanics models

Computational mechanics groups past histories into causal states when they induce the same conditional distribution over futures. Shalizi and Crutchfield show that the resulting ϵ -machine is minimal among equally predictive representations under their assumptions. This directly formalizes "finite state stores the effective structure of history and constrains future continuation," but probabilistically rather than geometrically [16].

5.7 Positional number systems and p-adic geometry

In base- b expansion, n digits identify an interval of length b^{-n} , up to the familiar dual-expansion endpoints. In the p -adic integers, n digits identify a congruence class modulo p^n , exactly a p -adic ball of radius p^{-n} . No experiment is needed to establish localization or continuation in these systems; it is built into their metric and algebra [17-19].

5.8 Symbolic control and finite abstraction

Control theory does not grant finite symbolic control models unconditionally. Under incremental stability, nonlinear systems can admit approximately bisimilar symbolic abstractions with arbitrarily selected precision; bounded state spaces can yield finite abstractions. These results show exactly which stability assumptions allow finite symbolic organization to represent continuous controlled behavior. Without such assumptions, reachability may be difficult or undecidable [20,21].

5.9 Undecidability and computational irreducibility

Universal computation imposes a hard boundary. Moore showed that low-dimensional dynamical systems can simulate Turing machines, making broad long-term questions undecidable even with exact initial conditions. Related hybrid-system results establish decidable subclasses and undecidable generalizations. Therefore no universal experiment or algorithm can map all finite constraints to all infinite outcomes across unrestricted recursive systems [22-25].

5.10 Cross-field synthesis table

Field	Finite object	Continuation	Established answer
Symbolic dynamics	Prefix cylinder	Infinite symbolic tail	Nestedness, entropy, coding
Automata	State / right-congruence class	Accepted finite or infinite continuation	Finite-state recognizability

Field	Finite object	Continuation	Established answer
IFS / fractals	Composed map word	Infinite address	Attractor cell, contraction, dimension
Information theory	Codeword / observed block	Source continuation	Information rate, typical cylinders
Domain theory	Compact finite approximant	Directed completion	Infinite object as supremum
Computational mechanics	Causal state	Future distribution	Minimal predictive state
Base-b / p-adics	Finite digit prefix	Digit tail	Exact interval or ultrametric ball
Symbolic control	Finite abstract state	Controlled trajectory	Approximate bisimulation under stability
Computability theory	Finite program/configuration	Unbounded computation	Undecidability boundary

6. Continued fractions as one special case

6.1 Exact cylinder geometry

For a regular continued fraction $x=[0;a_1,a_2,\dots]$, a finite word $w=(a_1,\dots,a_n)$ defines a cylinder containing all compatible infinite tails. If p_n/q_n and p_{n-1}/q_{n-1} are adjacent convergents, the cylinder endpoints are determined by these convergents, and its exact length is

$$|C_w| = 1 / [q_n(q_n + q_{n-1})].$$

Thus “focusing” is exactly denominator growth. The denominator q_n is an Euler continuant, so changing the digit order generally changes q_n and the width. This is the mathematical content of Experiment 1 [26-28].

6.2 Bounded alphabets

Restricting every partial quotient to $1 \leq a_i \leq K$ produces the classical compact set

$$E_K = \{ [0;a_1,a_2,\dots] : 1 \leq a_i \leq K \text{ for all } i \}.$$

For finite $K \geq 2$, E_K is a continued-fraction Cantor set with Hausdorff dimension strictly between zero and one. It has empty interior and permanent complementary gaps. Thermodynamic formalism and transfer operators determine its dimension, and rigorous algorithms compute dimensions with certified error bounds. This is the mathematical content of Experiment 2 [29-35].

6.3 Exact interpretation of project results

Project fact	Scientific status
12,295,602 enumerated words	A valid exact finite computation and implementation stress test.
Within-cell width dispersion	A descriptive statistic of non-injective coarse location versus continuant growth.
Order-only width dispersion	A finite sample of classical continuant order sensitivity.
$K=5$: 1229/4094 robust cells	An exact grid-aligned covering count for one K , depth, and resolution.
Depth 10 complete asymptotic mask	A finite saturation certificate for one target partition.
2865 missing cells structurally excluded	A finite-resolution manifestation of E_5 's complementary gaps.
245 greedy misses	A useful algorithmic benchmark, not a new theorem about continuation.

6.4 The correct generalization

The experiments do not support a new universal law derived from continued fractions. They provide a concrete instance of the standard coding theorem. The defensible general statement is:

General statement

For a compact admissible symbolic space with a continuous coding map, finite prefixes define nested compatible continuation sets; open-target reachability is intersection with the coded limit set; quantitative localization is controlled by the regularity and contraction of the coding.

7. What is genuinely universal - and what is not

7.1 Universality at the topological level

At the weakest topological level, the principle is extraordinarily general. Cantor space can continuously map onto every nonempty compact metric space. The standard coding theorem then guarantees prefix localization. This explains why the project repeatedly encountered the same structure in different mathematical examples.

7.2 Why this can become tautological

The same theorem also weakens any ontological claim. If arbitrary continuous codings are allowed, one can impose a prefix structure on almost any compact metric object. The existence of such a coding does not show that nature stores the object as a sequence, that the coding is dynamically privileged, or that the finite prefixes are causal. The theory becomes scientifically substantive only after the coding is fixed by independent structure: a generating partition, transition law, measurement process, physical control channel, or canonical arithmetic expansion.

7.3 Raw depth is not intrinsic

Take any binary coding and group symbols into nonoverlapping blocks of length two. The represented points and reachable set remain unchanged, but a prefix of $2n$ binary symbols becomes a prefix of n block symbols. Minimum depth is divided by approximately two. The alphabet size changes from 2 to 4. Reblocking by length b changes both quantities arbitrarily. Therefore:

raw prefix depth and alphabet size are properties of a representation, not of the represented object.

This counterexample does not require computation. It directly invalidates a universal interpretation of $K^*(J)$ and $N^*(J;K)$.

7.4 More defensible normalized quantities

- Information cost: $I(w) = -\log \mu([w])$ under a declared source or equilibrium measure.
- Geometric cost: $G(w) = -\log \text{diam}(C_w)$ under a declared metric.
- Computational cost: time or space on a fixed machine model and exact input encoding.
- Control energy or time: physical resources in a specified dynamical system.
- Residual freedom: conditional entropy, measure, or dimension after a prefix.

These quantities can still change under general coordinate transformations. A serious universality claim must specify an allowed equivalence class, such as bi-Lipschitz, bi-Hölder, bounded-block, or computable conjugacy, and prove stability within that class.

7.5 Localization is not forward predictability

A generative prefix can narrow a coded object while a forward dynamical system amplifies uncertainty. Positive Lyapunov exponents make initially nearby states separate exponentially. Entropy measures the rate at which new information is required to maintain predictive resolution. Therefore the project's localization principle and the arrow of time are not the same theorem.

7.6 Universal computation blocks a universal control atlas

When the transition system can simulate a Turing machine, questions such as eventual reachability, recurrence, termination, or long-term behavior can encode the halting problem. Some restricted systems remain decidable, but no finite experimental campaign can establish a complete control-cost law for unrestricted recursive systems. The research program must declare a tractable model class.

8. Novelty audit of Experiments 1-3

8.1 Experiment 1

Claim tested: finite continued-fraction words can steer cylinder location and focus cylinder width, with order effects and incomplete coverage. The exact facts are sound project computations. The mechanism is classical: cylinders, convergents, determinant identities, continuants, and their rearrangements. Novelty verdict: no new foundational result. Potential residual value: a benchmark dataset for search algorithms and exact arithmetic implementations.

8.2 Experiment 2

Claim tested: persistent gaps arise from the bounded coefficient alphabet rather than insufficient depth. This is precisely the limit-set interpretation of restricted-digit continued fractions. The finite statement that depth 10 saturates one particular dyadic mask is a project certificate, but the qualitative conclusion is classical. Novelty verdict: no new foundational result. Potential residual value: an independently reproducible finite saturation certificate.

8.3 Proposed Experiment 3

Quantity	Meaning	Prior framework	Verdict
$K^*(J)$	Minimum resource-family index whose limit set intersects J	Nested attractor/language threshold	Representation-specific; not intrinsically novel
$N^*(J;K)$	Shortest prefix whose continuation set lies in J	Shortest cylinder containment / coding depth / stopping set	Raw depth fails recoding invariance
$O^*(J;K)$	Extra geometric precision beyond target width	Scale overshoot / Moran stopping cover / renewal-adjacent	Most defensible, still coordinate-dependent
Pareto(K,N,O)	Nondominated resource triples	Multiobjective coding/control	Packaging alone is not a theorem
Greedy regret	Difference from exact optimum	Search and approximation algorithms	Potential algorithmic benchmark

8.4 The decisive defect in the original Experiment 3 design

Release blocker

The proposed full run treats K and raw word length as if they measure intrinsic finite organization. They do not. Block coding changes both without changing the represented outcomes. A large exact table would therefore map the chosen notation and grammar as much as the underlying geometry.

The correct response is not to improve the enumeration. It is to redesign the quantities before enumeration.

8.5 What can legitimately be claimed now

- The project independently reproduced standard continued-fraction coding behavior with exact arithmetic.
- The implementation can certify finite target containment, saturation, and greedy-policy failures.
- The cross-disciplinary qualitative principle is an established theorem under compact continuous coding.
- No evidence currently supports a new universal law or a coding-invariant control-cost geometry.
- A narrow quantitative theorem may remain available after normalization and a focused prior-art audit.

9. Research questions that remain defensible

The following are not declared open problems. They are precise candidate questions not resolved by the primary sources reviewed here in the exact target-conditioned form. Each requires a specialist bibliography search before novelty is asserted.

9.1 Candidate A: recoding-stable control cost

Let two symbolic codings represent the same compact set and be related by a bounded-block or bi-Hölder conjugacy. Determine which normalized target-containment costs differ by at most an additive or multiplicative constant. Raw depth fails immediately; geometric information may be stable under stronger equivalences.

Question A: When is $\inf\{-\log \text{diam}(C_w) : C_w \subseteq U\}$ invariant up to $O(1)$?

A theorem here would separate intrinsic geometry from arbitrary grammar. A counterexample would be equally valuable because it would show that the proposed resource geometry cannot be universal.

9.2 Candidate B: target-clearance law for overshoot

For a compact attractor A and target interval J , define its attractor clearance

$$\Delta_A(J) = \sup\{\text{dist}(x, J^c) : x \in A \cap J\}.$$

This is the radius of the largest target-centered neighborhood around an attractor point that remains inside J . The candidate law is that, for a finite conformal IFS with separation and bounded distortion, the optimal cylinder overshoot is controlled up to a system constant by the ratio $|J|/\Delta_A(J)$:

$$O^*(J) = \log_2(|J|/\Delta_A(J)) + O(1).$$

If true, this would explain why targets near gaps or boundaries appear expensive without invoking a new control substance. It would reduce “resource geometry” to established attractor geometry plus a bounded coding term. If false, the residual identifies the additional invariant that is missing.

9.3 Candidate C: target-conditioned minimum-depth law

For dyadic cells $J_m(x)$ containing a typical attractor point x , investigate whether a normalized minimum containment depth converges to a constant determined by a Lyapunov exponent or pressure. Ordinary orbit stopping and renewal theorems are adjacent, but N^* optimizes over all cylinders in a spatial target, not only the prefix orbit of a preselected point. The distinction must be checked carefully against shrinking-target and Moran-cover literature.

9.4 Candidate D: exact decision complexity

Fix a finite rational Möbius IFS and rational interval J . Classify the complexity of deciding whether a cylinder of length at most n lies in J , and of computing the minimum geometric overshoot. Bounded-depth search is finite, but a meaningful result requires complexity in the bit length of n , endpoints, and map coefficients. Runtime plots cannot prove a complexity class; formal reductions and certified algorithms are required.

9.5 Priority ranking

Priority	Question	Value	Reason
1	Target-clearance law	High	Geometric, falsifiable, directly explains project observations
2	Recoding-stable normalized cost	High	Necessary for any universality claim
3	Target-conditioned depth distribution	Medium	May overlap shrinking-target/renewal theory
4	Exact decision complexity	Medium	Potentially strong but requires theoretical CS expertise
5	Large Pareto heat map	Low	Descriptive and representation-dependent without prior theorem

10. Exact theorem-first experimental program

Rule

No experiment should ask whether the standard coding theorem is true in another familiar system. That work is already done. Computation is justified only when it distinguishes a precise theorem from a precise counterexample.

10.1 Gate 0 - novelty and theorem design

1. Search the exact terms “Moran cover,” “stopping set,” “cylinder containment,” “shrinking target,” “inradius,” “bounded distortion,” “continued-fraction Cantor set,” “target interval,” and “overshoot” in MathSciNet, Zentralblatt, and citation graphs of Hutchinson, Mauldin-Urbański, Hensley, and Sinai-Ulcigrai.
2. Write one theorem statement with assumptions, quantified constants, and a falsifying counterexample form.
3. Prove the theorem for a uniform self-similar benchmark before touching continued fractions.
4. Freeze the metric, target convention, cylinder convention, and permitted recodings.
5. Proceed to computation only if the theorem is not already in the literature and the analytical benchmark does not settle it.

10.2 Immediate experiment R1 - Target clearance versus optimal overshoot

This is the only immediate computational experiment recommended by this paper, and only after Gate 0.

Protocol field	Frozen specification
Objective	Test whether $O^*(J) - \log_2(J /\Delta_A(J))$ remains bounded across scale.
Systems	Middle-third Cantor set as analytic benchmark; continued-fraction E_K for $K=2,3,5$.
Targets	All interior dyadic cells at $m=8, \dots, 12$ for the pilot; central domain reported separately from boundary cells.
Arithmetic	Exact rational endpoints for self-similar benchmark; arbitrary-precision integer Möbius arithmetic and certified interval bounds for E_K .
Primary estimand	$R_K(J) = O^*(J) - \log_2(J /\Delta_A(J))$.
Primary hypothesis	There exists C_K independent of m and J such that $ R_K(J) \leq C_K$ for all reachable pilot cells.
Falsifier	The maximum $ R_K(J) $ grows systematically with m after exact certification, or a single certified counterexample violates the proposed bound.
Expansion rule	Increase to $m=16$ only if the pilot leaves a genuine asymptotic ambiguity and the theorem remains unsolved.
Publication-grade output	A theorem or counterexample; the heat map is secondary evidence only.

10.3 Required certificates and controls

- Exact containment by cross multiplication; floating-point values only for display.
- Independent implementations of cylinder generation and target containment.
- Reversed enumeration, clean-process, cache-off, and serialization tripwires.
- Published benchmark identities for the middle-third Cantor set and continued-fraction cylinder widths.
- Explicit distinction between exact attractor intersection, outer approximation, and unresolved interval bounds.
- No target-derived oracle, no sampled tails, and no inference from apparent saturation.
- Every reported maximum accompanied by a machine-readable witness word, target, and exact endpoints.

10.4 Experiment R2 - Recoding invariance audit

This is primarily a proof exercise with a small computational check.

Field	Specification
Construction	Represent the same attractor with original symbols, 2-block symbols, and 3-block symbols.
Expected exact result	Raw N^* scales approximately by 1, $1/2$, and $1/3$; alphabet size scales as $ \Sigma $, $ \Sigma ^2$, $ \Sigma ^3$.
Candidate invariant	Geometric cost G^* and information cost I^* , normalized to the same metric or measure.

Field	Specification
Theorem target	Bound the difference in normalized costs under bounded-block recoding.
Stop rule	If normalization does not remove arbitrary recoding distortion, terminate the universality claim for that quantity.

Because the failure of raw depth is already analytically obvious, this experiment must not become a large enumeration. Its purpose is to validate definitions and expose edge cases in a proposed invariant.

10.5 Deferred experiment R3 - Target-conditioned depth law

Run only after a dedicated shrinking-target and renewal-theory audit. Freeze a probability measure on the attractor, define whether targets are selected by measure or by grid, and distinguish the prefix of the sampled point from the best cylinder anywhere in the target. Enumerate all reachable cells for modest m rather than sampling. Test a predeclared normalization derived from the system's Lyapunov exponent. A trend without a theorem is not a result.

10.6 What must not be run

- Another larger depth sweep to confirm permanent bounded-digit gaps.
- A cross-system demonstration that prefixes localize in base expansions, p -adics, automata, or contracting IFSs.
- A full Experiment 3 Pareto atlas using raw K and N as universal resources.
- A dimension computation intended as novelty for classical E_K sets.
- A simulation of general recursive systems intended to establish universal decidability or controllability.

11. Decision and roadmap

11.1 Scientific answer to the original question

Answer

Established mathematics already answers the qualitative question. Finite information localizes infinite continuation when the infinite object is supplied through a compact continuous symbolic coding. The admissible language fixes what is possible; the coding regularity fixes how fast uncertainty shrinks; entropy and dimension quantify remaining freedom. This is not a new law of continued fractions and not a universal law of unrestricted reality.

11.2 Research pivot

The project should pivot from “discovering a general mechanism” to “identifying invariant quantitative structure within a declared model class.” The strongest candidate model class is finite conformal IFSs with separation and bounded distortion, including bounded-digit continued fractions as a nonuniform arithmetic example.

11.3 Roadmap

1. Archive Experiments 1 and 2 as verified computational reproductions of established theory.
2. Adopt the coding theorem in Section 4 as the theoretical foundation and cite the established fields rather than re-testing it.
3. Delete raw alphabet ceiling and raw word depth from any claim of universal control cost. Retain them only as implementation-specific resources.
4. Complete Gate 0 for the target-clearance law.
5. Attempt the analytic benchmark proof for a separated self-similar IFS.
6. Run Experiment R1 at $m=8, \dots, 12$ only if the literature and benchmark do not settle the question.
7. Promote the project to a research paper only when it contains a theorem, a certified counterexample, or a rigorously new algorithmic complexity result.

11.4 Final plain-English conclusion

A finite prefix narrows an infinite possibility only because a specified rule tells us what that prefix means. Mathematics already has a general theory of this. The remaining scientific problem is not whether narrowing happens. It is which measures

of the narrowing belong to the underlying system rather than to our chosen notation, and what exact laws govern those measures.

Appendix A. Project evidence

The project record reports the following exact computations. These are reproduced here to preserve the evidence while separating it from the established theory.

Item	Reported fact
Enumeration	K=3 through depth 10: 88,572 words; K=5 through depth 10: 12,207,030 words; total 12,295,602.
Targets	5,370 frozen dyadic target intervals evaluated exactly.
Integrity	0 recurrence-matrix mismatches; 0 determinant mismatches; 0 child-nesting mismatches; 87 tripwire checks with 0 failures; 4/4 source mutations detected.
Width dispersion	At n=10,m=12: median within-cell dispersion 8.356644 bits for K=3 and 16.452808 bits for K=5.
Order effect	996 multiset classes; median order-only width dispersion 1.139240 bits; maximum 3.081195 bits.
Reachability	K=5: 1229/4094 cells (30.020%); K=3: 421/4094 cells (10.283%).
Greedy policy	At K=5,m=12: 740 exact-minimum successes, 244 delayed successes, 245 reachable targets missed.
Experiment 2	At K=5,m=12, depth 10 certified the complete asymptotic mask; 2,865 cells were structurally excluded by the alphabet.

Source: project experimental record supplied in the conversation on 6 August 2026. These facts have internal audit support as described in that record, but they are not presented here as externally peer-reviewed results.

Appendix B. Proof details and counterexamples

B.1 Uniform localization

Let $d(s,t) = 2^{-\ell(s,t)}$, where ℓ is the length of the longest common prefix. Every length- n cylinder has diameter at most 2^{-n} . Given $\epsilon > 0$, uniform continuity of π supplies $\eta > 0$ such that $d(s,t) < \eta$ implies $d_X(\pi(s), \pi(t)) < \epsilon$. Choose n with $2^{-n} < \eta$. Then every nonempty C_w with $|w|=n$ has diameter below ϵ .

B.2 Reachability equivalence

If $C_w \subseteq U$ and C_w is nonempty, then $U \cap A$ is nonempty. Conversely, take $x \in U \cap A$ and $s \in Y$ with $\pi(s) = x$. Since U is open, continuity of π at s gives a prefix cylinder $[w]_Y$ containing s whose image lies in U . This is why the target must be open, or why a positive-margin condition must be imposed for closed targets.

B.3 Block-recoding counterexample

Let B_b group every b consecutive symbols into one block. The map B_b is a conjugacy between the original sequence and a block-coded sequence over alphabet Σ^b . A length- bn original prefix and a length- n block prefix specify the same continuation set. Therefore any raw minimum-depth law can be rescaled by choosing b , while the represented object and geometric target remain unchanged. Alphabet size changes simultaneously from $|\Sigma|$ to $|\Sigma|^b$. This proves that raw depth and alphabet cardinality are not universal resources.

B.4 Coordinate distortion

Let $h: X \rightarrow X$ be a homeomorphism and replace π by $h \circ \pi$. Topological reachability is preserved after targets are transformed by h , but diameters can be distorted arbitrarily by a general homeomorphism. Bi-Lipschitz maps bound this distortion by constants; bi-Hölder maps change exponents. Any geometric universality claim must therefore name the allowed coordinate transformations.

B.5 Finite saturation does not imply global finite depth

For a fixed finite target partition, reachable cells have finitely many first-success depths, so their maximum is finite. This does not imply a uniform depth that works for every target at every resolution. As targets shrink, required depth generally grows. Experiment 2's depth-10 saturation is therefore a partition-specific certificate, not a global bound.

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[Hutchinson 1981](#)

[Mauldin and Williams 1988](#)

[Shannon 1948, Part I](#)

[Shannon 1948, Part II](#)

[Pola, Girard, and Tabuada 2008](#)

[Moore 1990](#)

[Jenkinson and Pollicott 2018 preprint copy](#)

[Rutar 2026 preprint](#)

Research integrity note. This document is a synthesis and novelty audit. The general coding theorem is presented as a standard consequence of established topology and symbolic dynamics, not as an original theorem. Candidate questions are explicitly not certified open. The proposed experiment is subordinate to a specialist prior-art search and an analytical benchmark proof.